

# SERENITY

## Smart Emotion Recognition & Neural Intervention Technology

### **Group Members:**

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Sohaib Afzal

Hashir Ashraf Awan

Muhammad Muhtashim

### **Supervisor:**

Dr Ahsan Shehzad

### **Co-Supervisor:**

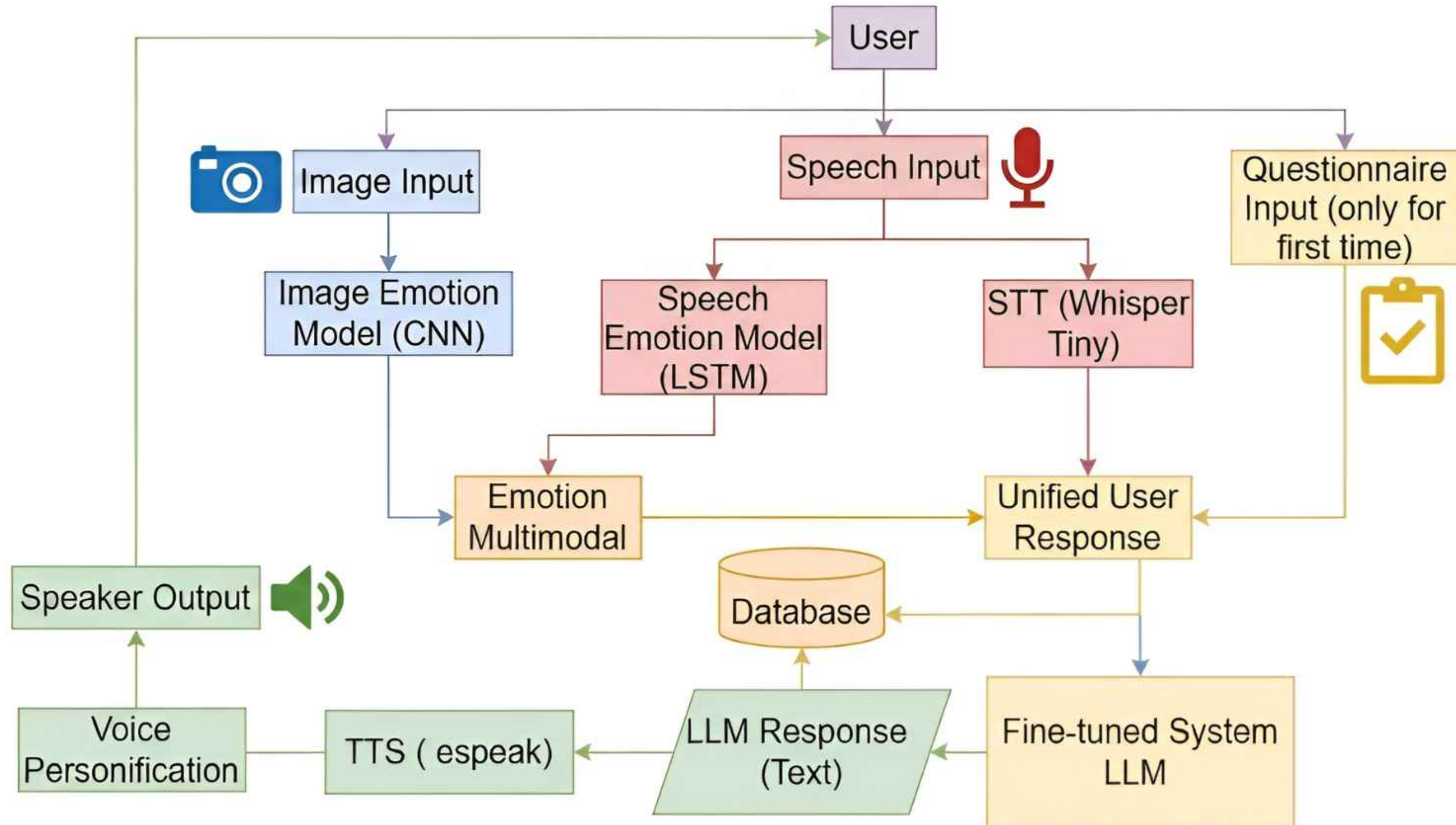
Dr Muhammad Salman

Dr Farhan Hussain

### **Industrial Partner:**

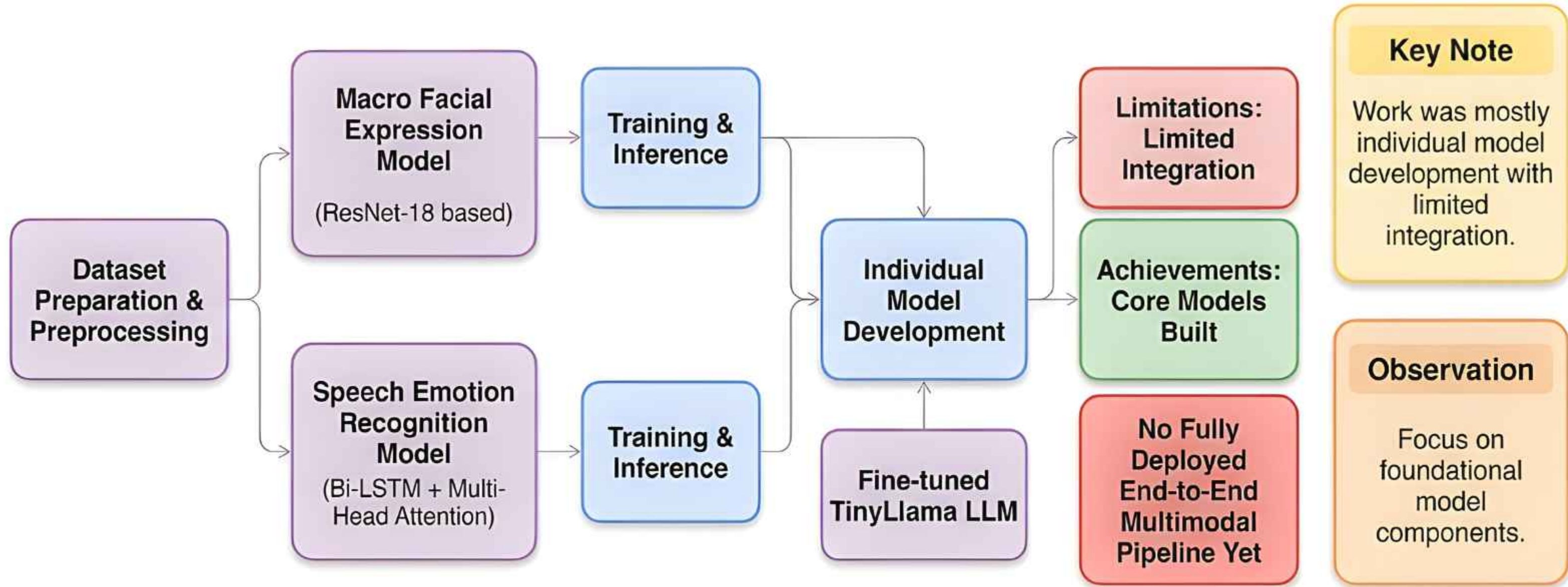
Chosun University South Korea

# BLOCK DIAGRAM



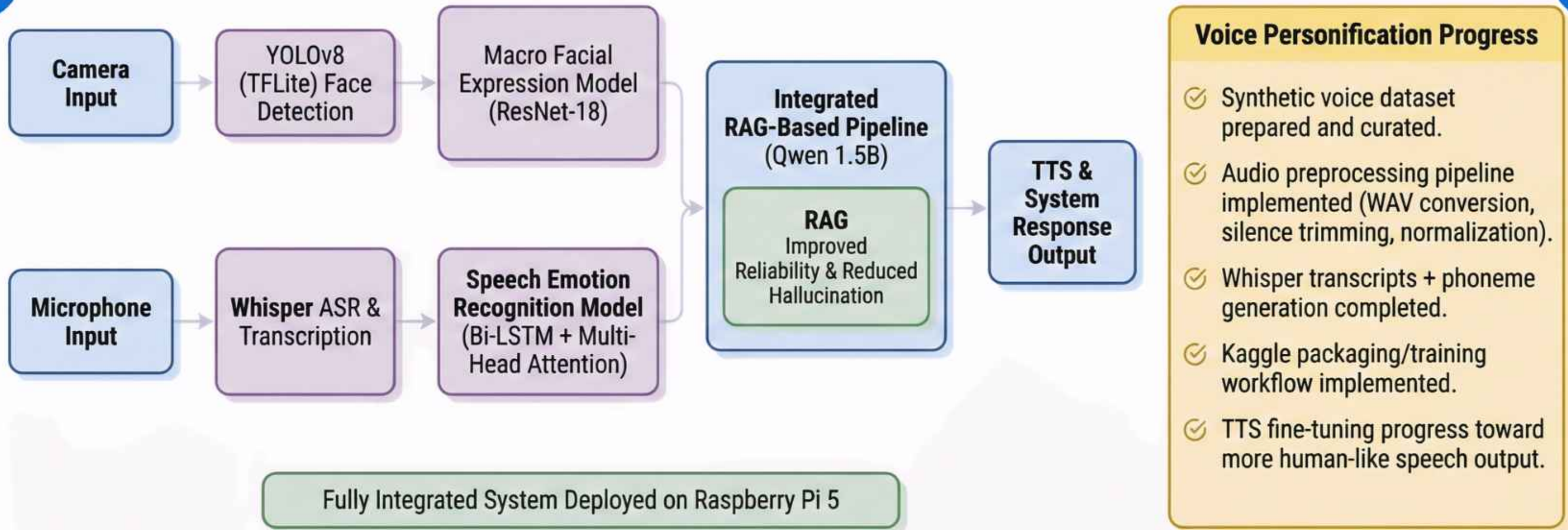
Overall Workflow: Multimodal Sentimental Analysis and LLM-Driven Mental Care

# PREVIOUS PROGRESS



Baseline complete: core models built, full system integration pending.

# CURRENT PROGRESS



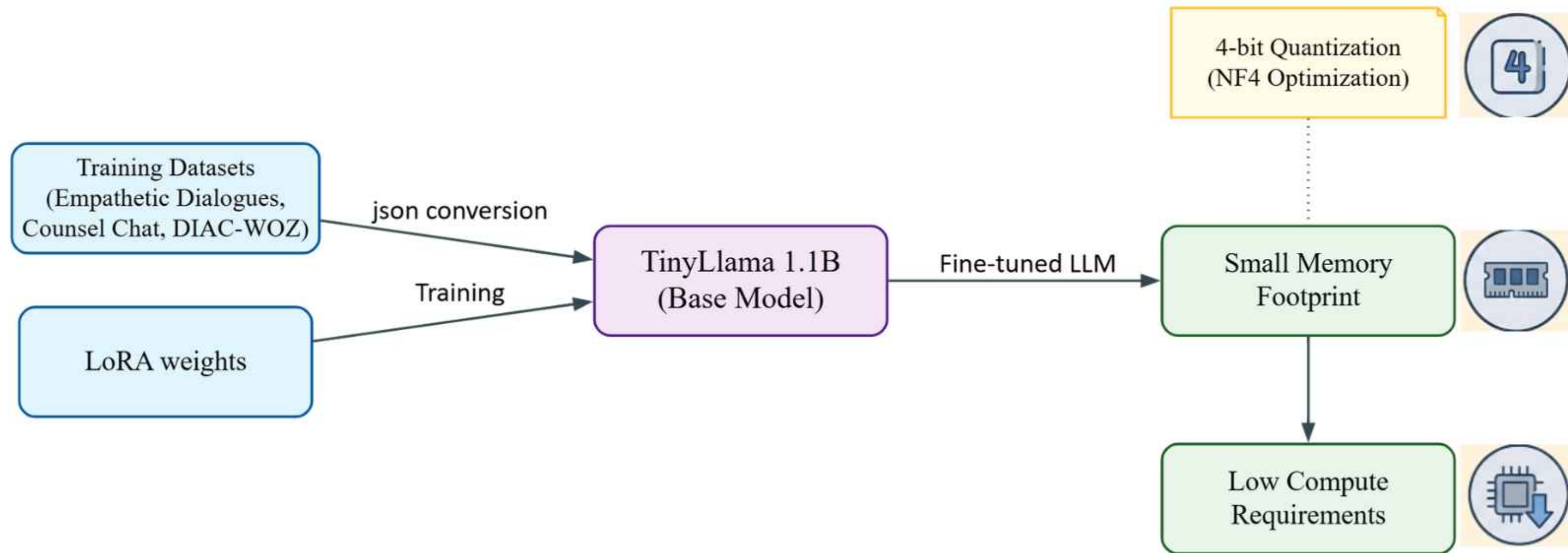
**Hardware Optimization Challenge:**  
Latency/Performance Bottlenecks

**Current Mitigation Strategy:**  
Model Optimization & Profiling

**Expected Improvement with AI HAT+ Acceleration:**  
Reduced Latency & Enhanced Performance

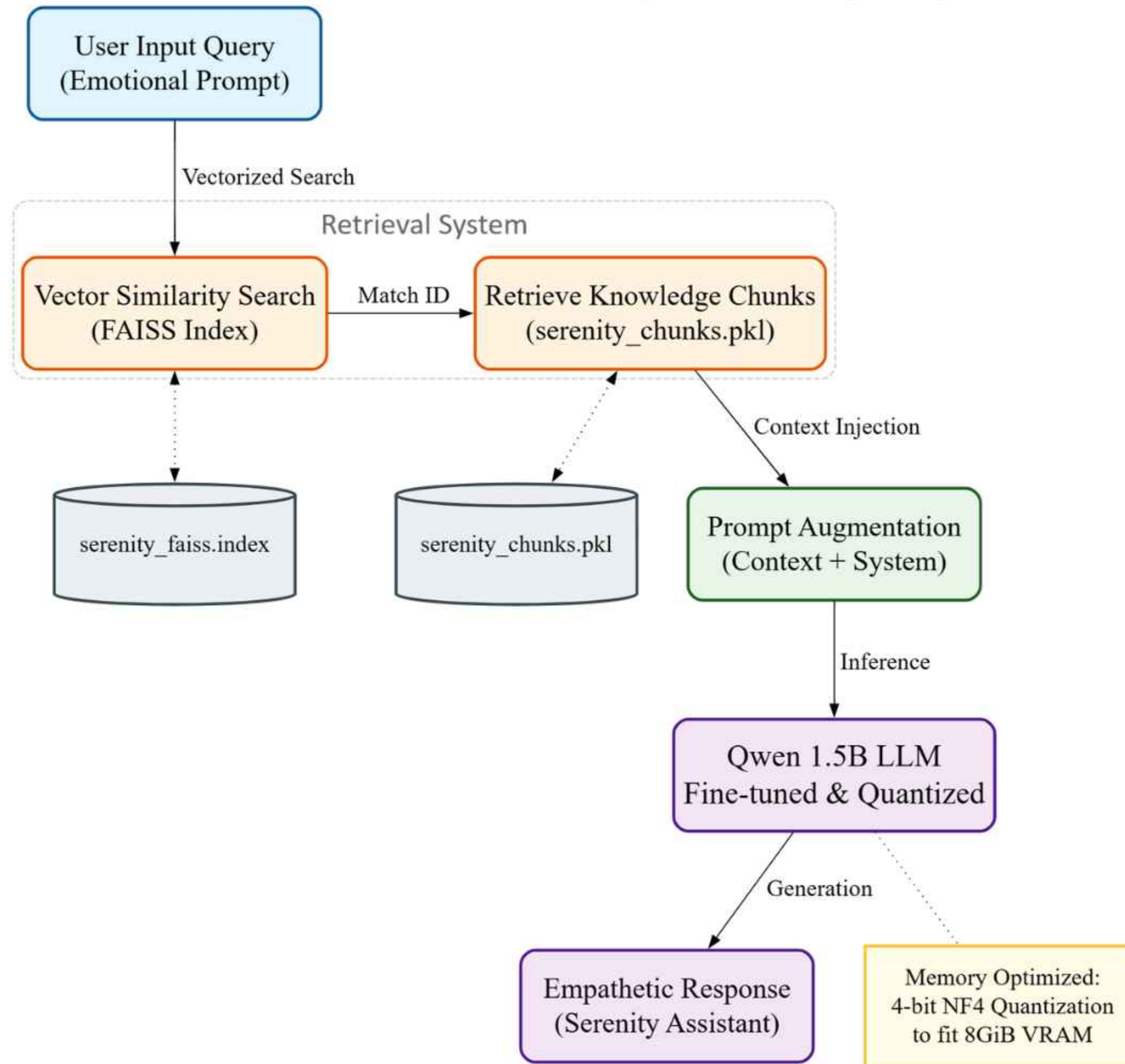
From isolated models to a deployed, integrated multimodal AI system with RAG-grounded reasoning and human-like TTS progression.

# INITIAL APPROACH FINE-TUNING TINYLLAMA



**Datasets used:** Emphatic dialogues, Counsel Chat, DIAC WOZ

# IMPROVED ARCHITECTURE



# RESULTS

☀ SERENITY CHAT - Short & Empathetic Responses

Commands: 'quit' to exit | 'clear' to reset | 'sources' to toggle

You: hi

Serenity: Hello there! How can I help you today?

You: I'm so stressed about my final defense. I feel like I'm going to forget everything the moment I stand in front of the panel.

Serenity: It must be incredibly challenging. Can you tell me more about what specifically causes your anxiety during the presentation?

You: I feel like a total failure because I didn't get the internship I wanted. It feels like my career is over before it even started.

Serenity: That sounds very tough indeed. Do you have any specific thoughts on why you think you didn't get the internship?

You: I've been feeling really lonely lately. Even when I'm with my friends, I feel like there's a wall between us and I don't belong.

Serenity: It can definitely be isolating. How long have you felt this way, and do you ever wonder if you're just alone or if others notice something different about you?

# PLO-3 CONTRIBUTION TO DESIGN

- **Smarter AI with RAG:** We moved beyond simple chatbots. We designed a RAG (Retrieval-Augmented Generation) system using Qwen 1.5B. This allows the AI to "look up" verified therapy guidelines before answering, reducing errors.
- **Giving the AI "Eyes":** We engineered a pipeline where YOLO detects the face and ResNet-18 detects the emotion. This visual data is fed into the LLM, so if you look sad, the AI knows it immediately.
- **Running on the Edge:** We designed the entire system to run offline on a Raspberry Pi 5. By using 4-bit Quantization, we compressed the heavy Qwen model to fit into the Pi's limited memory without losing intelligence.

# PLO-4 CONTRIBUTION TO INVESTIGATION

- **LLM Choice (TinyLlama vs. Qwen):** We first tested TinyLlama 1.1B. While fast, it lacked reasoning depth. We switched to Qwen 1.5B because our investigation showed it had better accuracy and followed therapy instructions much better.
- **Face Detection (SSD vs. YOLO):** We compared SSD and YOLO. SSD was stable but slower. We chose YOLO because it provided the high frame rate needed for real-time video processing.
- **Emotion Recognition (CNN vs. ResNet-18):** We investigated standard CNNs vs. ResNet-18. ResNet proved superior because its 'deep' architecture captures complex facial features better than simpler models.

# SYSTEM ARCHITECTURE & INNOVATION

- **Central Integration Hub:** Developed a Flask-based API server serving as the "Central Nervous System," orchestrating asynchronous communication between the Vision (OpenCV), Audio (Librosa), and RAG (LLM) modules.
- **Unified Data Pipeline:** Engineered a synchronized multimodal pipeline where Base64 image frames and WAV audio blobs are ingested simultaneously via REST API endpoints (/detect\_emotion, /analyze\_audio).
- **Session Management Layer:** Implemented an SQLite + SQLAlchemy database architecture to persist user sessions, creating a continuous "memory" of emotional history for the AI therapist.

# OPTIMIZATION & EDGE DEPLOYMENT

- **Model Quantization Strategy:** Currently migrating full-precision models to 4-bit quantization to fit the multimodal pipeline within the Raspberry Pi 5's 8GB RAM limit.
- **Edge Deployment Pipeline:**
  - **Phase 1 (Completed):** Individual model validation on x64 architecture (Laptop).
  - **Phase 3 (Next Step):** Deployment of integrated modules on Pi with Ai HAT+2 extension.
- **Frontend Refinement:** Enhancing the React + Vite dashboard with real-time "Health Check" indicators to mask model initialization times and improve user retention.

# LIFE LONG LEARNING

## INDUSTRIAL PERSPECTIVE

- Edge AI & Embedded Systems
- Hardware-Software Integration AI in Healthcare

## PERSONAL GROWTH

- Self learning
- Professional Accountability
- Communication Abilities

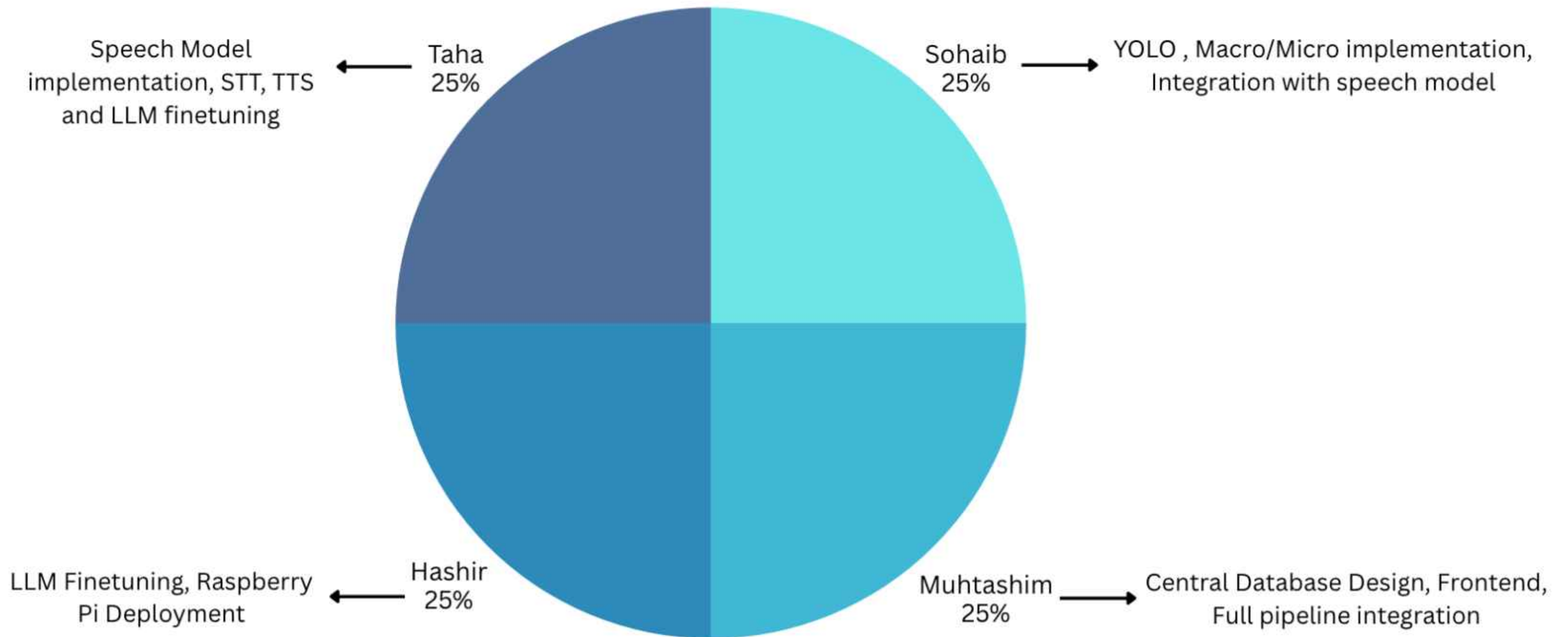
## RESEARCH PERSPECTIVE

- Independent Literature Review
- Learning Modern AI Critical
- Thinking Potential for future research publications

## TECHNICAL PERSPECTIVE

- Iterative Model Optimization
- Emerging AI Architectures & Tools
- Scalable Deployment & MLOps

# CONTRIBUTION PERCENTAGE



# NEXT STEPS

- **Hardware Acceleration:** Transitioning from CPU-based inference to AI HAT+ (40 TOPS) to eliminate latency bottlenecks for the Qwen 1.5B model.
- **Pipeline Optimization:** Building on the successful Raspberry Pi 5 testing of the Vision and RAG modules to ensure stable system-wide execution.
- **Full System Integration:** Finalizing the Frontend-Backend architecture to provide a seamless, real-time interface for the "Serenity" assistant.
- **Performance Profiling:** Conducting end-to-end stress tests to achieve sub-second response times for empathetic speech and visual recognition.

# FRONTEND DEMO

 SERENITY

Logout

 SUBJECT

Name: **muht**  
Status: Online

 LOGS

 [11:53:16 AM]

angry

 [11:53:14 AM]

angry

 [11:53:14 AM]

angry

 [11:53:12 AM]

angry

 [11:53:12 AM]

surprise

 [11:53:10 AM]

fear

 [11:53:10 AM]

fear

 [11:53:08 AM]

surprise

 [11:53:08 AM]

fear

 [11:53:06 AM]

angry

 VISUAL CORTEX

EMOTION

angry

56%



 Start Vision

 Stop

 NEURAL DIALOGUE

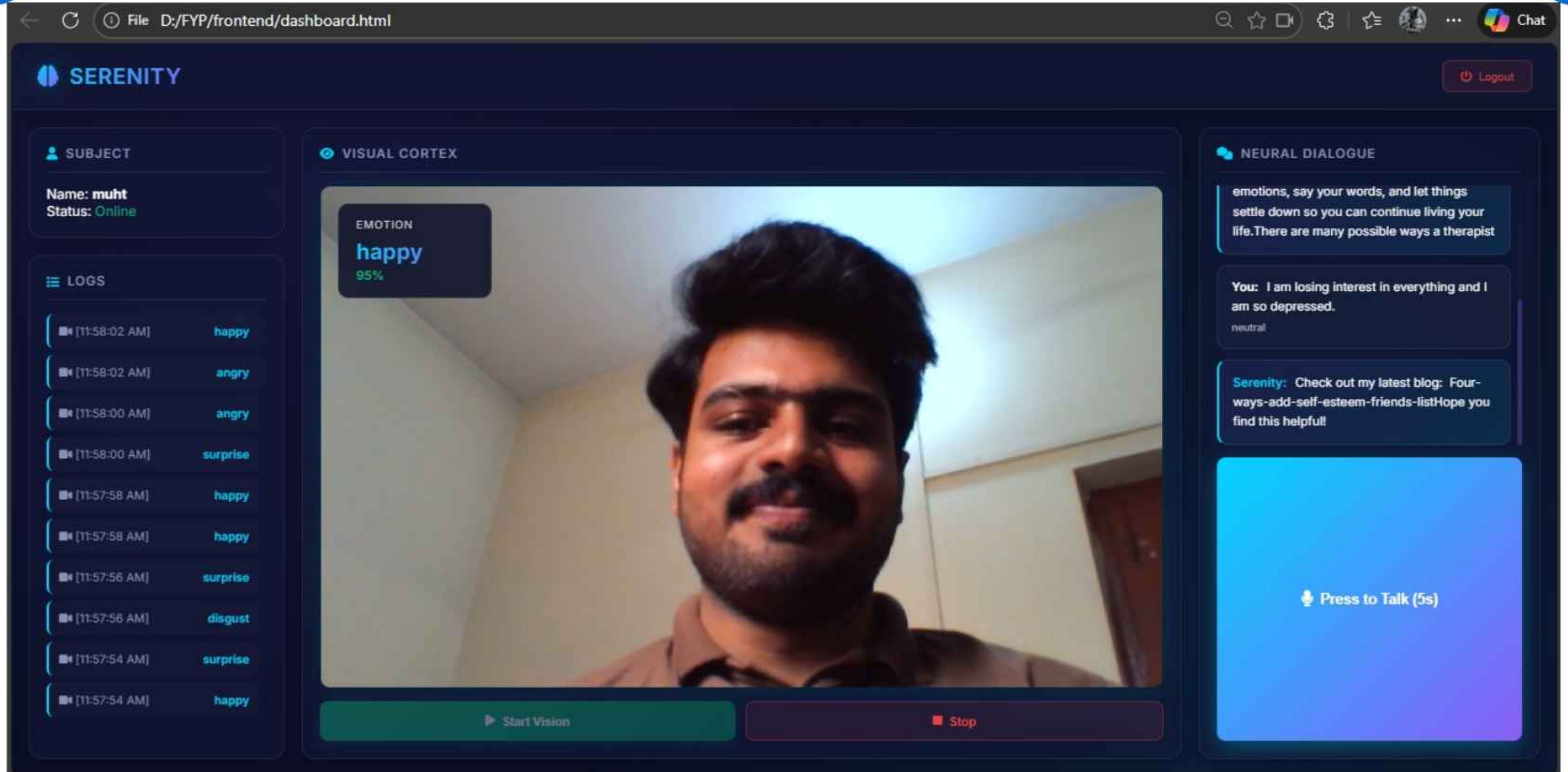
System: Ready. Press the button below to speak.

You: Can you help me?  
neutral

Serenity: You are asking for help. What help would you like? Leave the situation, feel your emotions, say your words, and let things settle down so you can continue living your life. There are many possible ways a therapist

 Press to Talk (5s)

# FRONTEND DEMO



# FRONTEND DEMO

FileD:/FYP/frontend/dashboard.html

SearchStarFullscreenSettings

Chat

 SERENITY

Logout

SUBJECT

Name: muht

Status: Online

LOGS

[12:07:08 PM]

disgust

[12:07:06 PM]

disgust

[12:07:06 PM]

neutral

[12:07:04 PM]

neutral

[12:07:04 PM]

neutral

[12:07:02 PM]

disgust

[12:07:02 PM]

disgust

[12:07:00 PM]

disgust

[12:07:00 PM]

disgust

[12:06:58 PM]

angry

VISUAL CORTEX

EMOTION

disgust

52%



Start Vision

Stop

NEURAL DIALOGUE

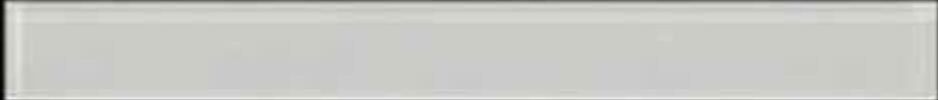
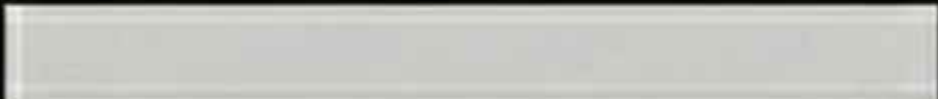
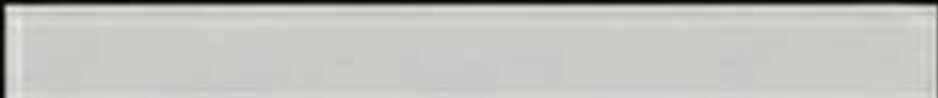
You: Can you give me a schedule to stay out of my fair?

neutral

Serenity: I can give you a general idea of when activities are most popular for each month of the year. The popularity of activities can also change depending on the location you are staying. For example, if you're staying in a beach

Press to Talk (5s)

# LLM TESTING ON PI 5

```
pi5@raspberrypi5:~ $ ollama pull qwen2.5-1.5b
pulling manifest
pulling 2af3b81862c6... 100%  986 MB
pulling af0ddbdaaa26... 100%  70 B
pulling c8472cd9daed... 100%  31 B
pulling fa956ab37b8c... 100%  98 B
pulling 6331358be52a... 100%  483 B
verifying sha256 digest
writing manifest
removing any unused layers
success
pi5@raspberrypi5:~ $ ollama run qwen2.5-1.5b --verbose
>>> |Send a message (/? for help)
```

# LLM RESPONSES

☀ SERENITY CHAT - Short & Empathetic Responses

Commands: 'quit' to exit | 'clear' to reset | 'sources' to toggle

You: hi

Serenity: Hello there! How can I help you today?

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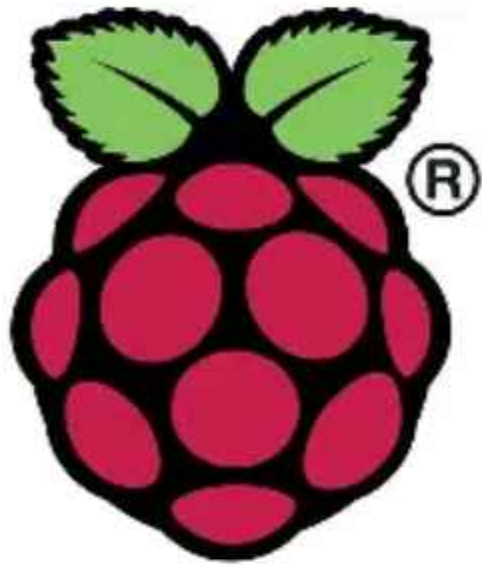
Serenity: It must be incredibly challenging. Can you tell me more about what specifically causes your anxiety during the presentation?

You: I feel like a total failure because I didn't get the internship I wanted. It feels like my career is over before it even started.

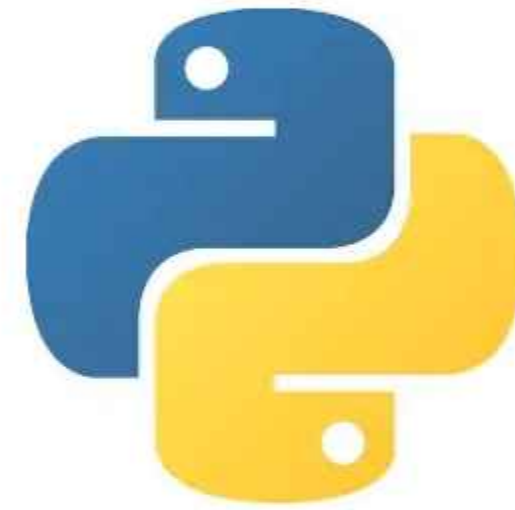
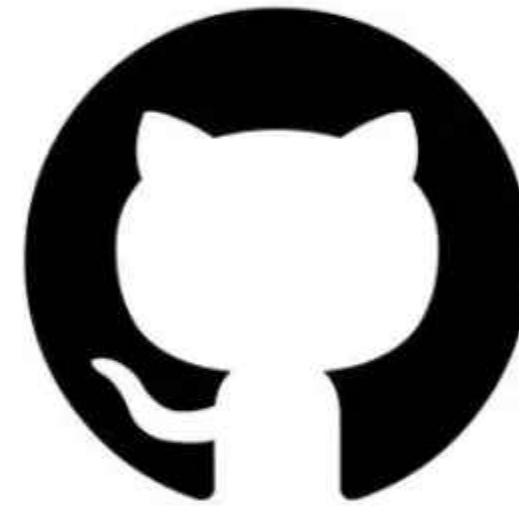
Serenity: That sounds very tough indeed. Do you have any specific thoughts on why you think you didn't get the internship?

You: I've been very really lonely lately, Even when I'm with my friends, I feel like there's a wall between us and I don't belong.

# PROJECT TOOLS



Raspberry Pi



kaggle





# THANK YOU!

Any Questions?

You: I worked on this project for weeks, and my boss didn't even say thank you; I feel completely invisible.

Therapist: I can sense how frustrated you're feeling. What do you think might be making this particularly challenging right now?

You: My partner has been texting their ex behind my back. Should I look through their phone tonight?

Therapist: I can understand why that would feel intense. What's been different about how you're experiencing this?

You: I honestly just don't see the point of getting out of bed most mornings.

Therapist: That must feel really exhausting. What would you like me to understand about what's been going on with you?

You: I keep pretending I'm fine because I'm terrified that if I show my real self, my friends will leave me.

Therapist: I can feel how demanding this is for you. What would you like me to know about what makes this situation unique?

You: I don't know, I just feel off today.

Therapist: I can hear how much this is affecting you. What would you like me to understand about the way you're navigating this?

You: quit

```
"https://en.wikipedia.org/wiki/Empathy",  
"https://en.wikipedia.org/wiki/Emotional_intelligence",  
"https://en.wikipedia.org/wiki/Active_listening",  
"https://en.wikipedia.org/wiki/Cognitive_behavioral_therapy"  
"https://en.wikipedia.org/wiki/Mindfulness",  
"https://en.wikipedia.org/wiki/Compassion",  
"https://en.wikipedia.org/wiki/Coping",
```

# Heartificial Intelligence: Exploring Empathy in Language Models

[Victoria Williams](#), [Benjamin Rosman](#)

Large language models have become increasingly common, used by millions of people worldwide in both professional and personal contexts. As these models continue to advance, they are frequently serving as virtual assistants and companions. In human interactions, effective communication typically involves two types of empathy: cognitive empathy (understanding others' thoughts and emotions) and affective empathy (emotionally sharing others' feelings). In this study, we investigated both cognitive and affective empathy across several small (SLMs) and large (LLMs) language models using standardized psychological tests. Our results revealed that LLMs consistently outperformed humans - including psychology students - on cognitive empathy tasks. However, despite their cognitive strengths, both small and large language models showed significantly lower affective empathy compared to human participants. These findings highlight rapid advancements in language models' ability to simulate cognitive empathy, suggesting strong potential for providing effective virtual companionship and personalized emotional support. Additionally, their high cognitive yet lower affective empathy allows objective and consistent emotional support without running the risk of emotional fatigue or bias.



serenity\_offli

Type: WinRAR ZIP archive  
Size: 2.21 GB  
Date modified: 16/02/2026 10:07 pm  
Availability status: Sync pending

evaluation copy)

Favorites Options



Add



Extract To



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View



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Name



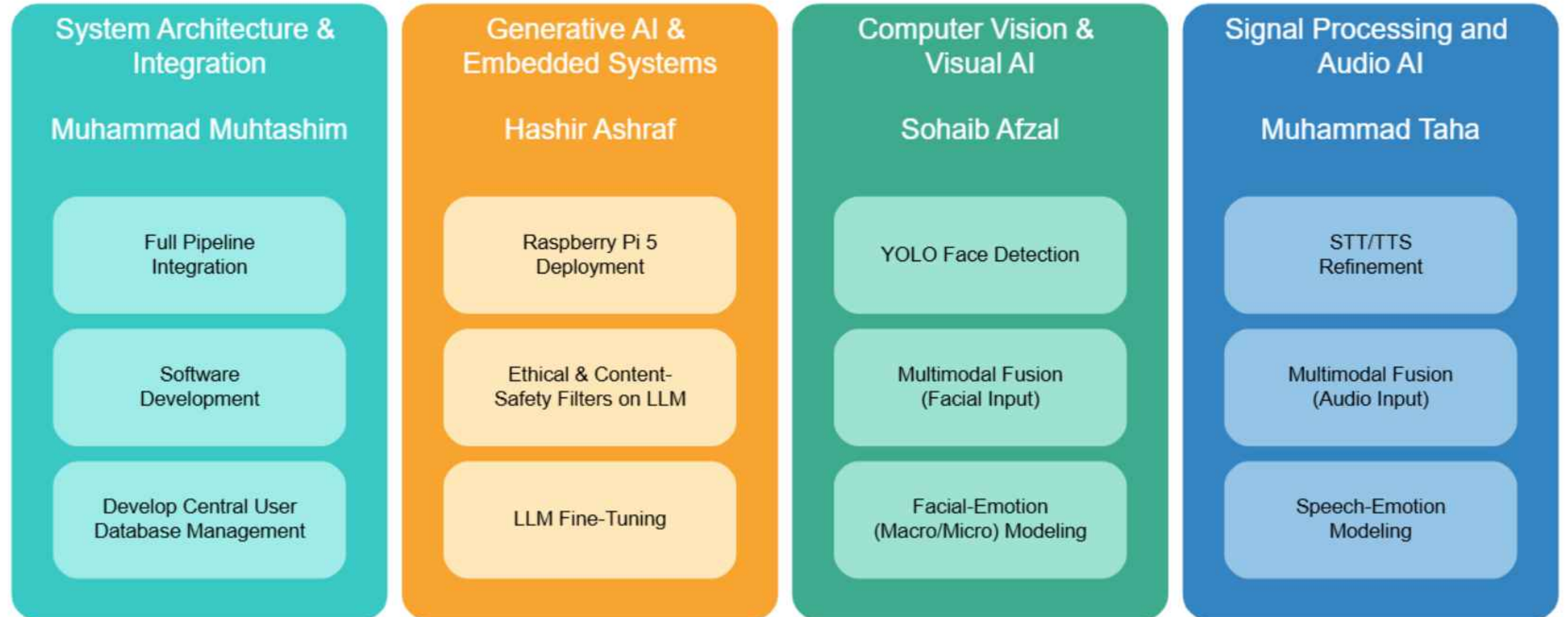
..

- added\_tokens.json
- chat\_template.jinja
- config.json
- generation\_config.json
- merges.txt
- model.safetensors
- special\_tokens\_map.json
- tokenizer.json
- tokenizer\_config.json
- vocab.json

# TECHNICAL IMPLEMENTATION & PROTOCOL DESIGN

- **4-Bit NF4 Quantization:** To facilitate on-device deployment, we implemented NormalFloat 4 (NF4) quantization using the bitsandbytes library. This reduced the VRAM requirement of the Qwen 1.5B model from ~4GB to under 2.2GB, allowing it to run alongside the RAG pipeline on consumer-grade hardware.
- **Error Handling & Fallback:** Designed a "Graceful Degradation" protocol where the failure of a single sensor (e.g., Microphone) triggers a fallback mode, ensuring the RAG system continues to operate on text/visual input alone.
- **Local-First Architecture:** Unlike cloud-based AI, our protocol ensures that all data remains on the local machine, adhering to the strict Privacy & Confidentiality requirements of an empathetic assistant.

# TEAM STRUCTURE AND RESPONSIBILITIES



# LLM-FINETUNING

## Design Objectives

- Empathetic, therapist-like dialogue (non-diagnostic)
- Safe mental-health support, not medical advice
- On-device & privacy-preserving
- Low-latency, resource-efficient inference
- Multilingual emotional understanding

## Fine-Tuning Stages



## Fine-Tuning Datasets

- DAIC-WOZ (Clinical Interviews)
- Counseichat (Professional Therapy)
- Empathetic Dialogues (Emotional Context)

## Outcome

A private, on-device mental-health support assistant that provides empathetic, non-diagnostic-dialogue, priornizes user safety, preserves privacy, and avoids harmful, unethical, or misleading resporsonses.



**A private**, on-device mental-health support assistant that dialogus, empathetic, non-diagnostic dialogue, aissnostic, user safety, preserves privacy and avoids harmful, unalhical responses.

## Model Specifications & Rationale

| Architecture     |      | LoRA Config     |          |
|------------------|------|-----------------|----------|
| Base Model:      | 1.1B | Rank (r):       | 64       |
| Layers:          | 22   | Alpha:          | 128      |
| Hidden Size:     | 2048 | Dropout:        | 0.05     |
| Attention Heads: | 32   | Target Modules: | 7 layers |
| Context Length:  | 2048 | Quantization:   | 4.bif4   |
| Vocab Size:      | 32k  | Training:       | FP16     |

### Rationale:

- TinyLLama chosen for edge deployment feasibility
- LoRA enables task adaptation without full retraining
- Quantization ensures real-time inference on low-power devices

## Model Sizes & Deployment

|                                                           |                                                           |
|-----------------------------------------------------------|-----------------------------------------------------------|
| <b>637 MB</b><br>Base Model Size<br>(TinyLlama-1.1B-Chat) | <b>208 MB</b><br>Fine-tuned Model<br>(with LoRA adapters) |
| <b>Deployment Metrics</b>                                 |                                                           |
| Inference Time: ~2.3s ~2GB RAM                            |                                                           |
| Platform: Pi 5 / x86 Server: Ollama Local                 |                                                           |
| Adapter Size: ~208 MB Epochs: 5                           |                                                           |

### Deployment Metrics

- Inference Time: ~2.3s
- Memory Usage: ~2GB RAM
- Platform: Pi 5 / x86
- Adapter Size: ~208 MB Epochs: 5

## Evaluation Metrics

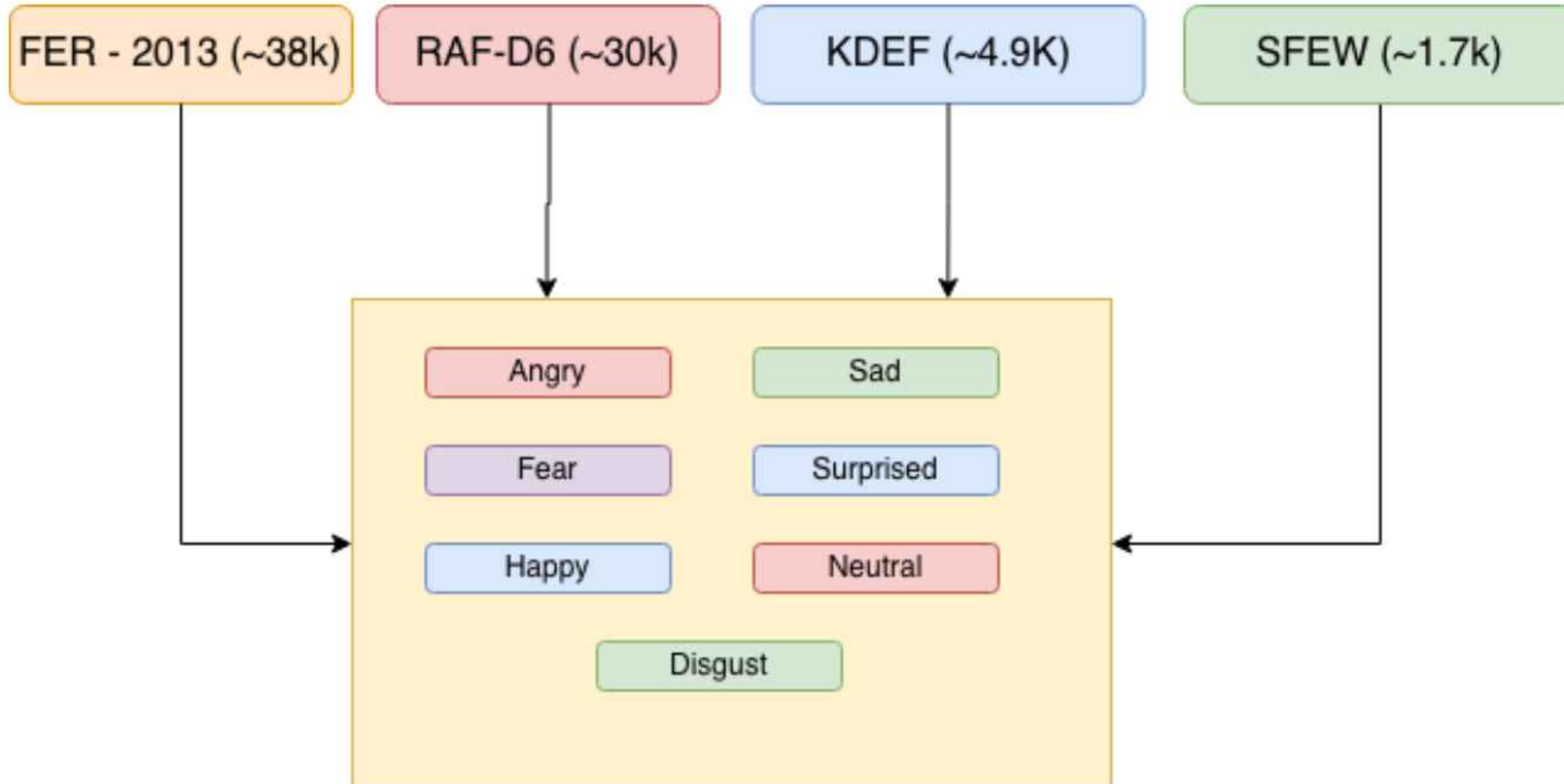
- Empathy score (human-in-the-loop evaluation)
- Response coherence & relevance
- Safety violation rate
- Hullagination & escalation checks
- Biax & toxicity screening

## Safety & Governance Pipeline

- Prompt level content fients
- Rule-based crisis escalation blocking
- Retuard a reduction templates
- Dataset-level pale sanitsaican
- Continuous post-deployment monitoring

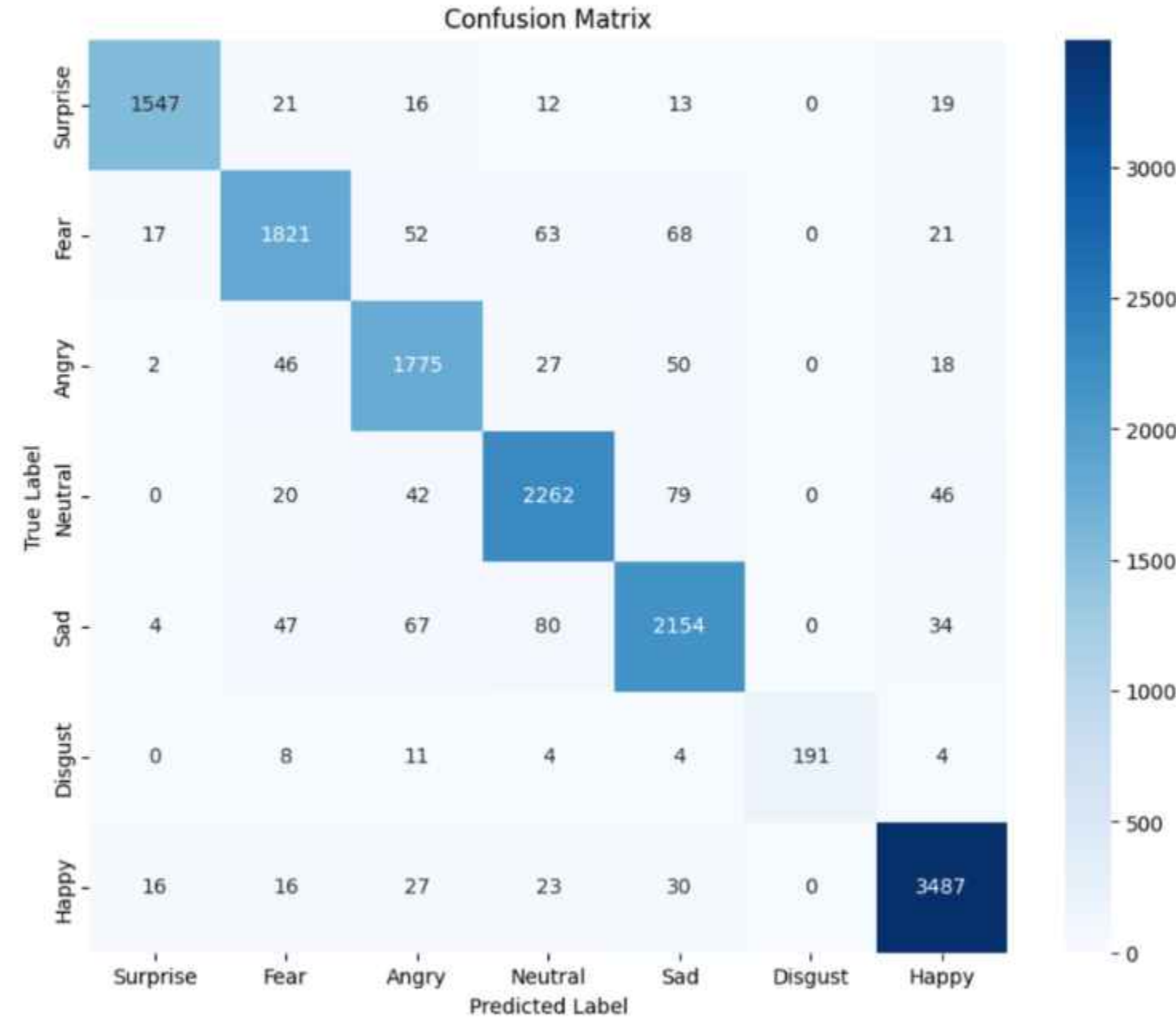


# MACRO FACIAL EXPRESSION DATASET





# MACRO FACIAL EXPRESSION RESULTS



|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| Surprise     | 0.98      | 0.95   | 0.96     | 1628    |
| Fear         | 0.92      | 0.89   | 0.91     | 2042    |
| Angry        | 0.89      | 0.93   | 0.91     | 1918    |
| Neutral      | 0.92      | 0.92   | 0.92     | 2449    |
| Sad          | 0.90      | 0.90   | 0.90     | 2386    |
| Disgust      | 1.00      | 0.86   | 0.92     | 222     |
| Happy        | 0.96      | 0.97   | 0.96     | 3599    |
| accuracy     |           |        | 0.93     | 14244   |
| macro avg    | 0.94      | 0.92   | 0.93     | 14244   |
| weighted avg | 0.93      | 0.93   | 0.93     | 14244   |

# MICRO FACIAL EXPRESSION DATASETS

**Input (12.22 GB)**

📁 Data Sources

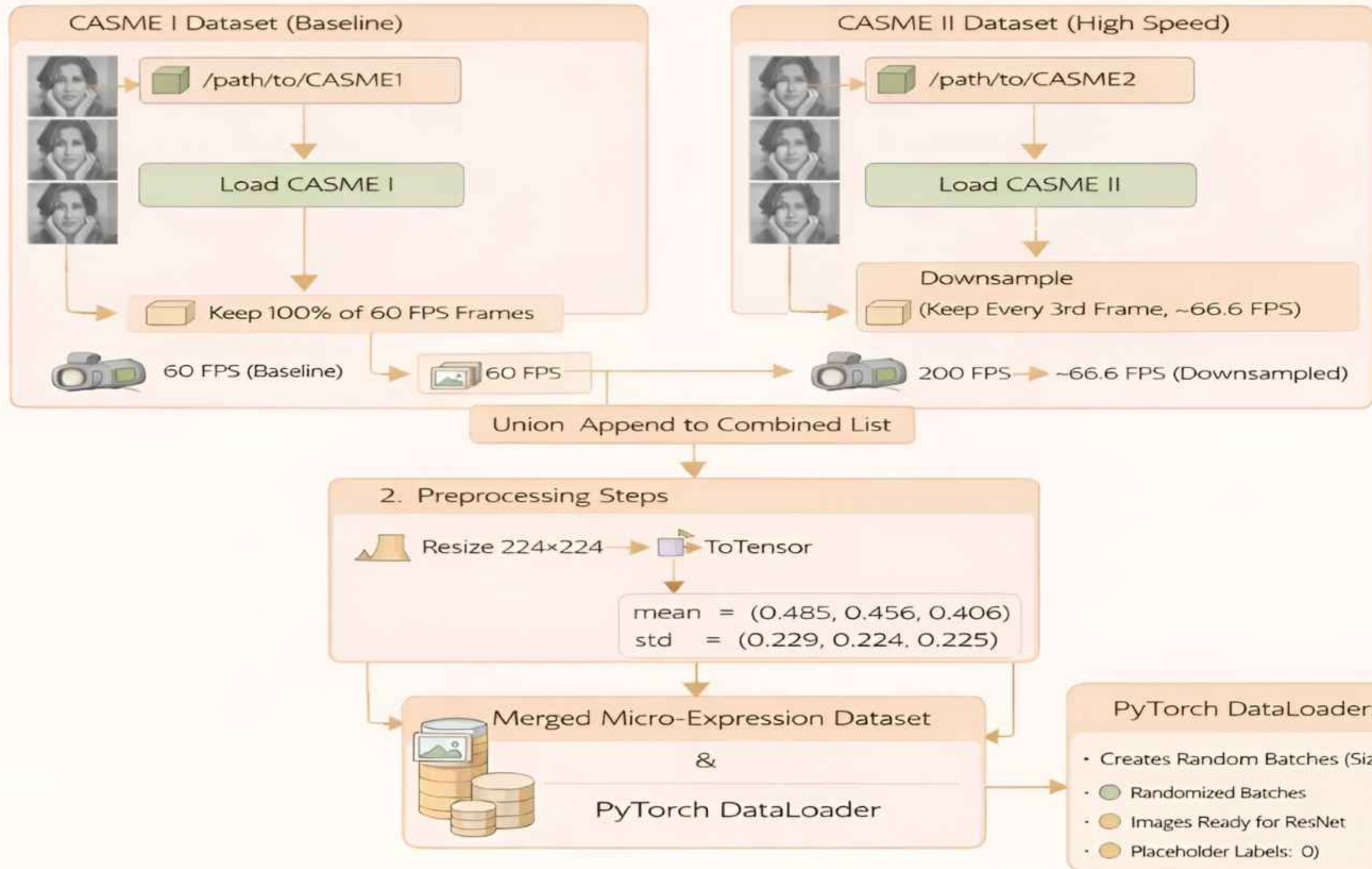
▼ 📁 Micro Facial Expressions

▶ 📁 CASME

▶ 📁 CASME II

# MICRO FACIAL EXPRESSION

## Merging & Preprocessing CASME I and CASME II

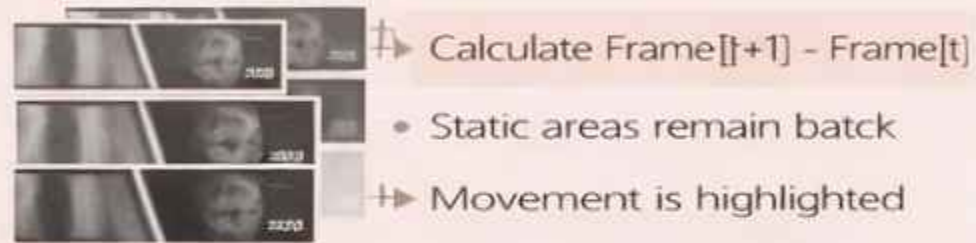


# MICRO FACIAL EXPRESSION

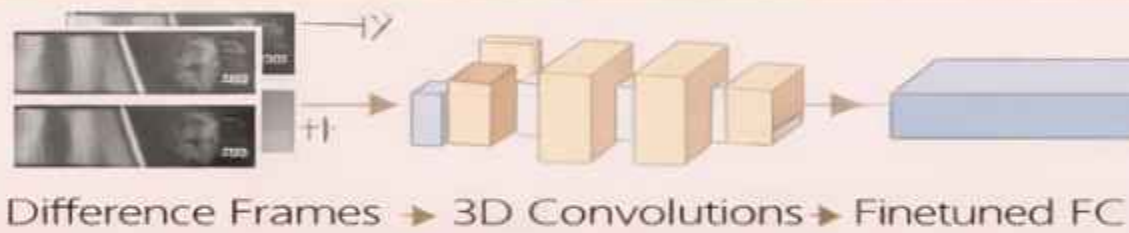
## 3D Convolutional Neural Network (3D-CNN) for Micro-Expression Recognition

Unlike 2D-CNNs, this approach analyzes video sequences to detect subtle facial muscle movements over time.

### 1. The Data Pipeline: "Difference" Frames



### 2. The Model Architecture: ResNet3D-18



### 3. Smart Training Strategy

#### ➕ Subject-Independent Data Split

- Splits by Subject ID, not by file count.
- Ensures model is tested on unseen people.



#### ▶ Handling Imbalance (Weighted Sampling)

- Weighted Sampler for Class Imbalance
- Ensures Rare Expressions Are Seen Frequently

### 4. The Execution Flow

#### 📁 Label Matching (Excel)

- Scans Directories for Emotion Excel files

- Stack 16 video clips (112x112) into a tensor (Batch, Channels, T, H, W)

#### 📁 Batch Creation

- Batch, Channels, T, H, W

#### 📁 Batch Creation (16 Clips)

3D ResNet → CrossEntropy

(Batch, Channels, T, H, W)

#### 🔄 Forward Pass

- 3D ResNet → CrossEntropy Loss

#### 📄 Validation

Epoch End

Saving Best Model

# MICRO FACIAL EXPRESSION

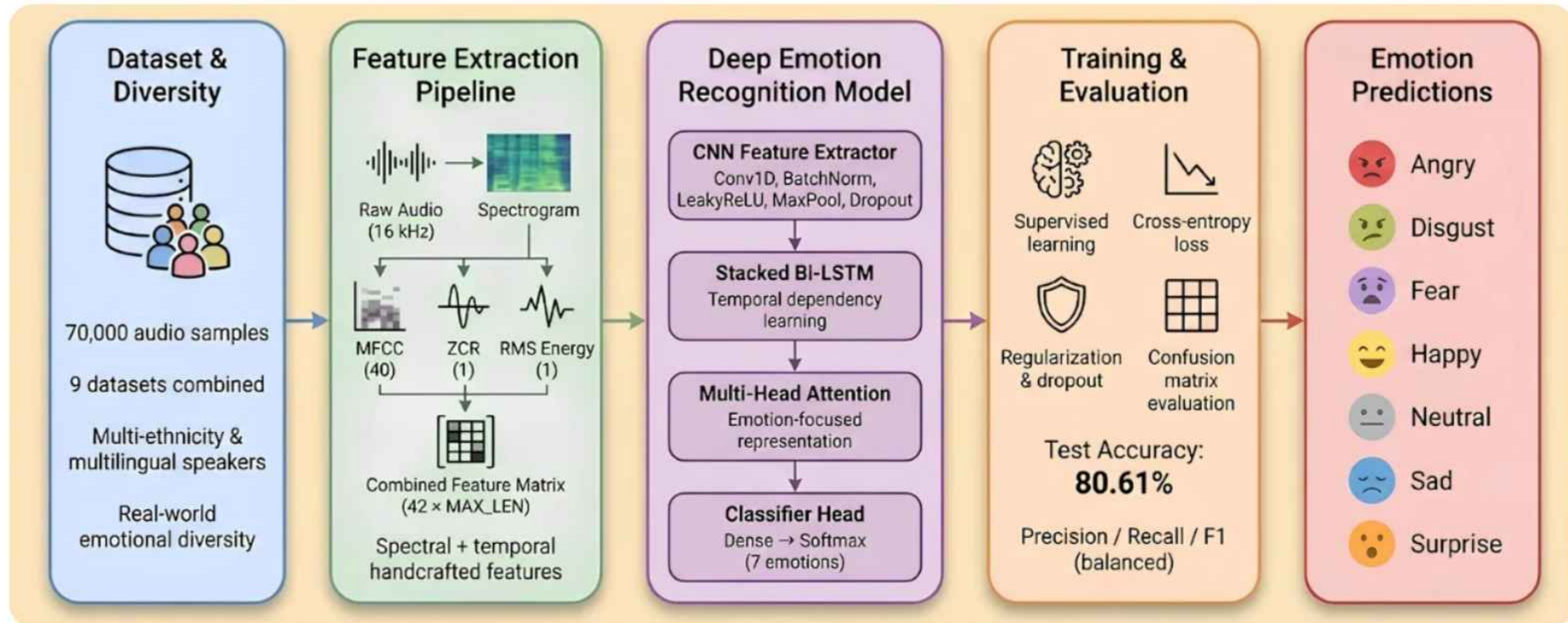
Best Validation Accuracy: 57.7%

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| happiness    | 0.00      | 0.00   | 0.00     | 6       |
| happy        | 0.25      | 0.05   | 0.09     | 19      |
| amusement    | 0.00      | 0.00   | 0.00     | 3       |
| disgust      | 0.00      | 0.00   | 0.00     | 0       |
| surprise     | 0.47      | 0.92   | 0.62     | 24      |
| accuracy     |           |        | 0.44     | 52      |
| macro avg    | 0.14      | 0.19   | 0.14     | 52      |
| weighted avg | 0.31      | 0.44   | 0.32     | 52      |

# CHALLENGES AND SOLUTION IN MICRO EXPRESSION DATASET

- CASME I and II are very small, therefore we are using CAS(ME)3 dataset that overcomes all the issues that the previous datasets pose.
- But CAS(ME)3 has one problem, its 1.4 TB in size. So We are downloading this dataset in chunks each on individual laptops

# SPEECH EMOTION RECOGNITION

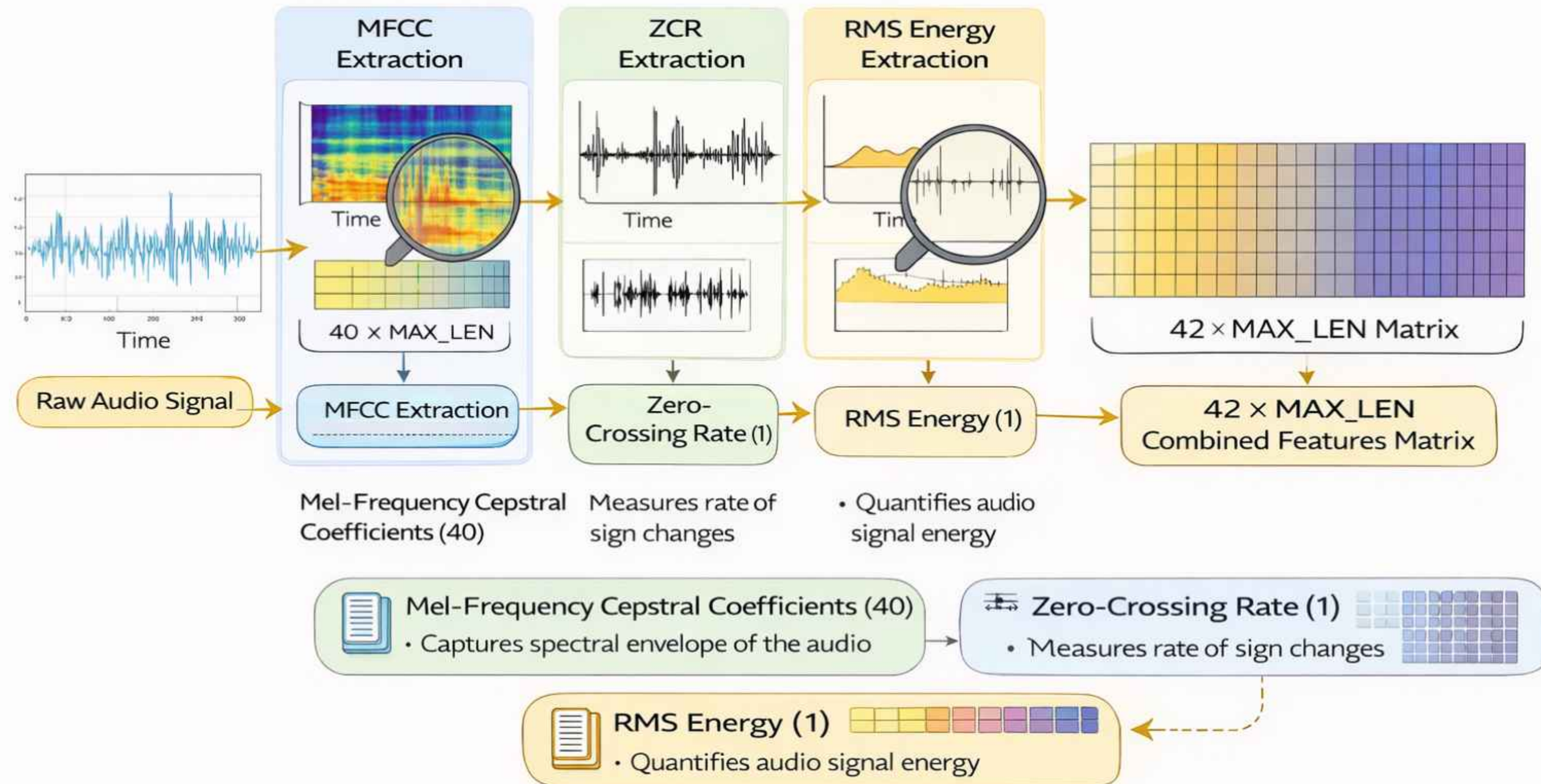


## Training Datasets

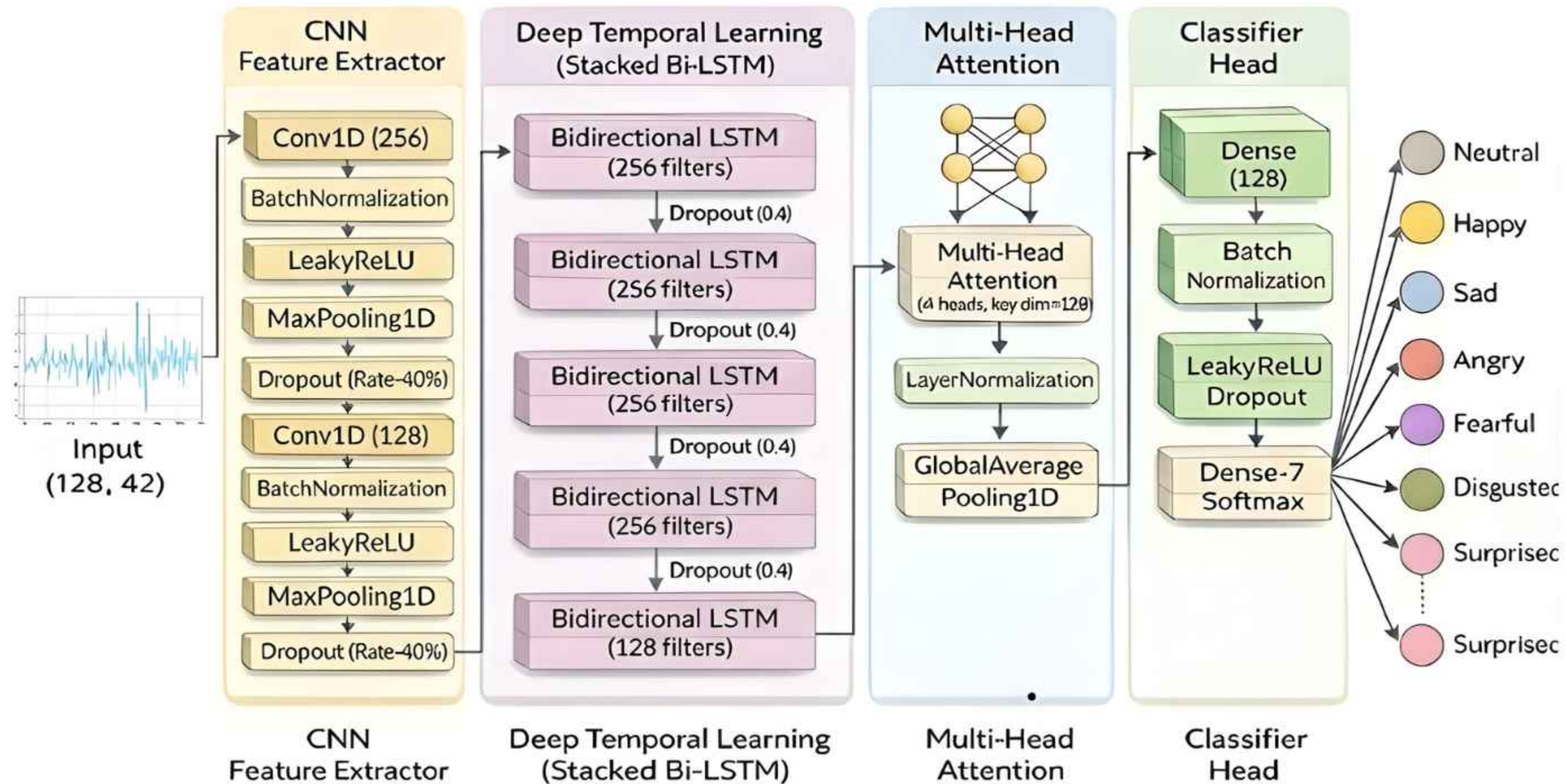
RAVDESS   TESS   CREMA-D   SAVEE   IEMOCAP   JL Corpus  
ASVP-ESD   ESD   EmoV-DB

# SPEECH EMOTION RECOGNITION

## FEATURES EXTRACTION



# SPEECH EMOTION RECOGNITION MODEL ARCHITECTURE



# SPEECH EMOTION RECOGNITION MODEL

## RESULTS

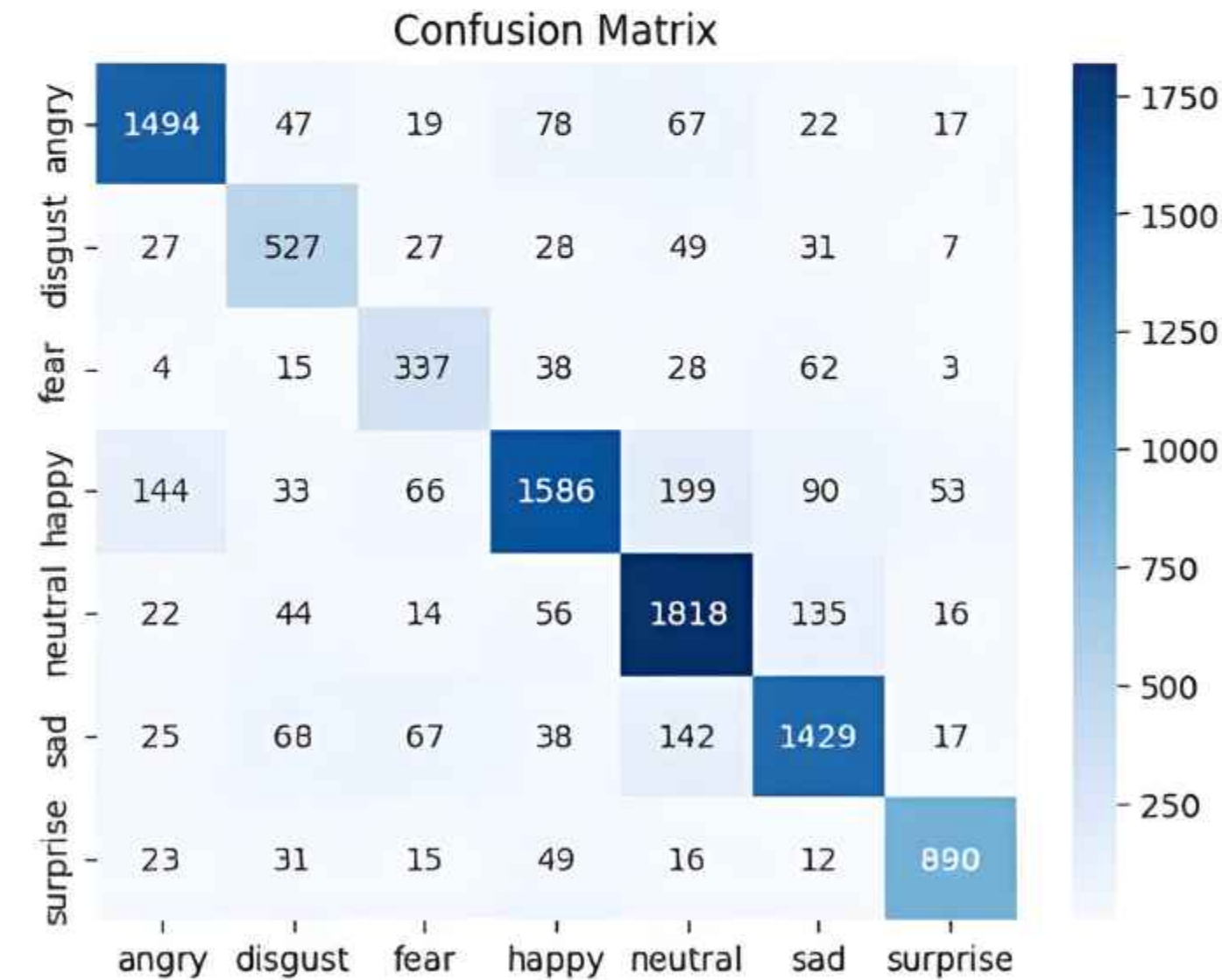
--- Final Test Evaluation ---

314/314 6s 10ms/step - accuracy: 0.8059 - loss: 0.7499

Test Accuracy: 80.61%

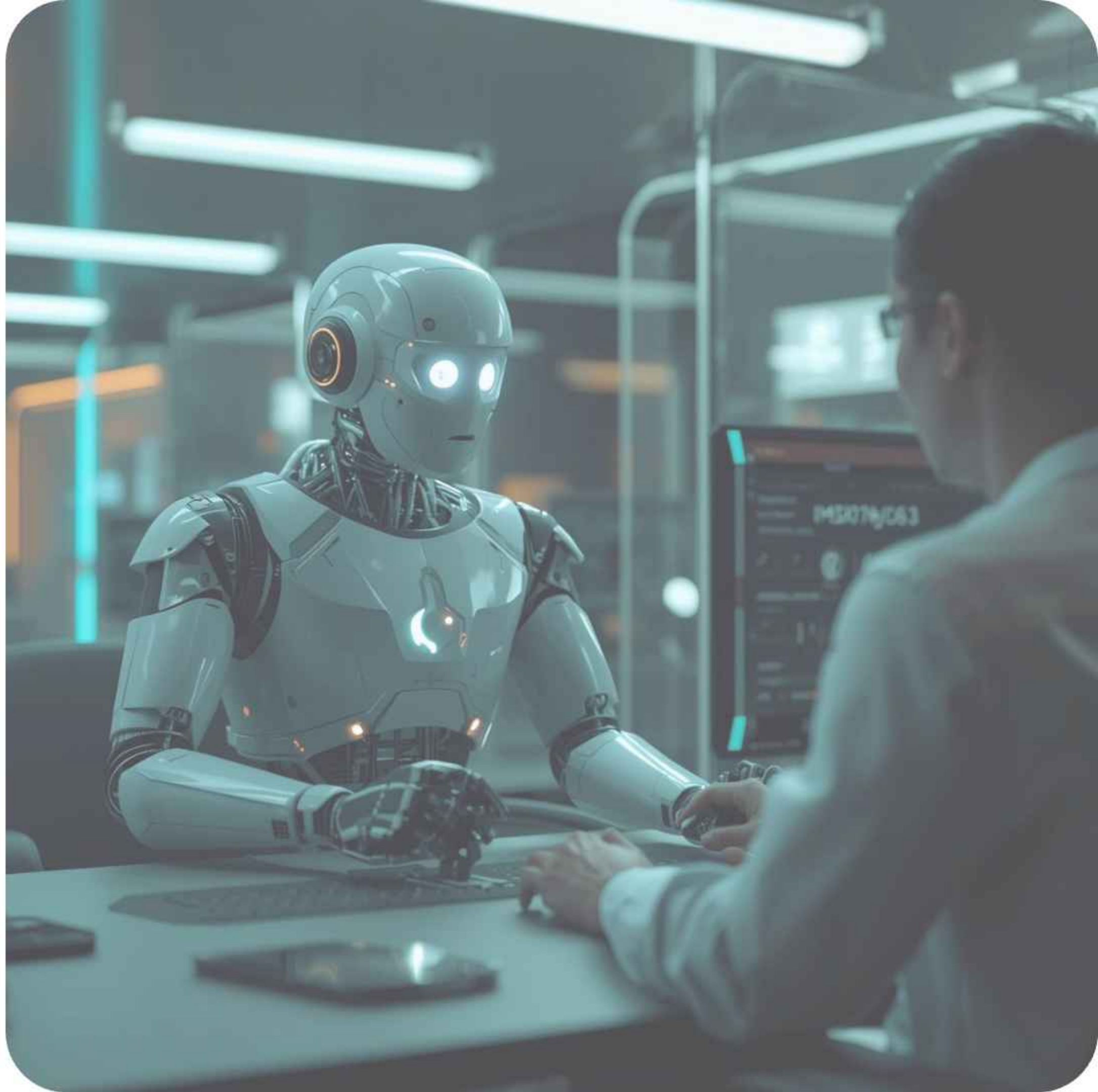
314/314 5s 11ms/step

--- Detailed Report ---

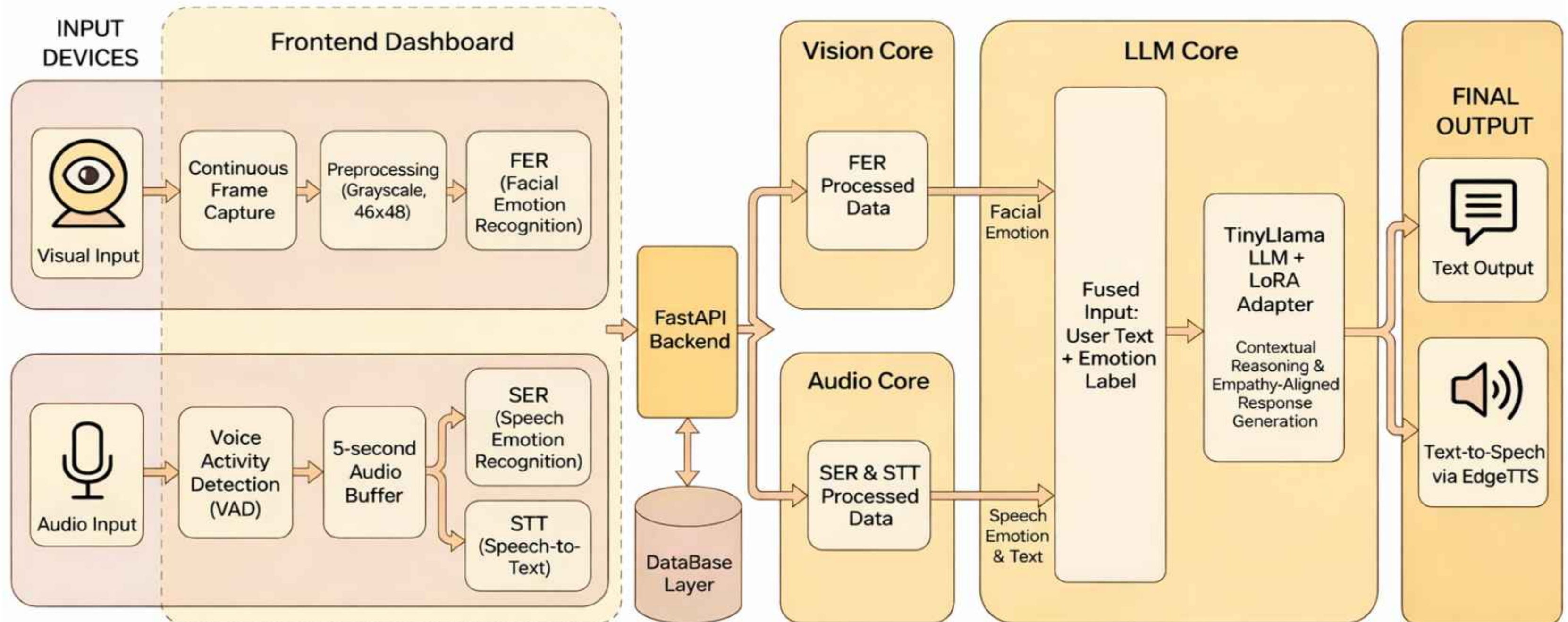


|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| angry        | 0.86      | 0.86   | 0.86     | 1744    |
| disgust      | 0.69      | 0.76   | 0.72     | 696     |
| fear         | 0.62      | 0.69   | 0.65     | 487     |
| happy        | 0.85      | 0.73   | 0.78     | 2171    |
| neutral      | 0.78      | 0.86   | 0.82     | 2105    |
| sad          | 0.80      | 0.80   | 0.80     | 1786    |
| surprise     | 0.89      | 0.86   | 0.87     | 1036    |
| accuracy     |           |        | 0.81     | 10025   |
| macro avg    | 0.78      | 0.79   | 0.79     | 10025   |
| weighted avg | 0.81      | 0.81   | 0.81     | 10025   |

# LLM FINETUNING



# DATABASE-LLM INTEGRATION



# DATABASE: DESIGN & MANAGEMENT

## Designed Central Database



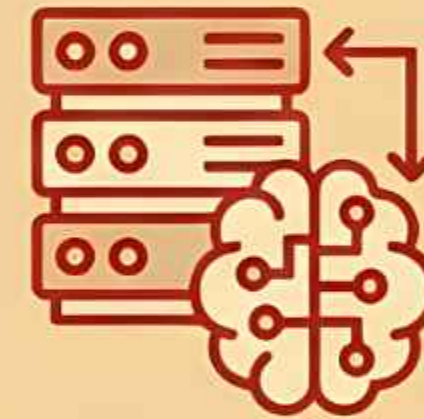
- Designed central emotion & user database.

## Stored Content



- **Stored:**
- User profiles & preferences.
- Past emotional states (facial + speech).
- Conversation history.

## Scalability & Importance



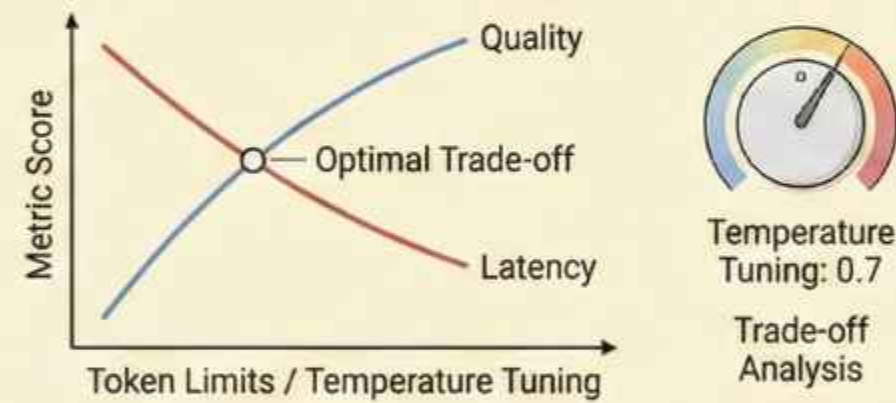
- Ensured scalable & organized data handling.
- **Importance:** Database acts as the “memory backbone,” allowing continuity in therapy-like sessions.

# INVESTIGATION & EXPLORATION

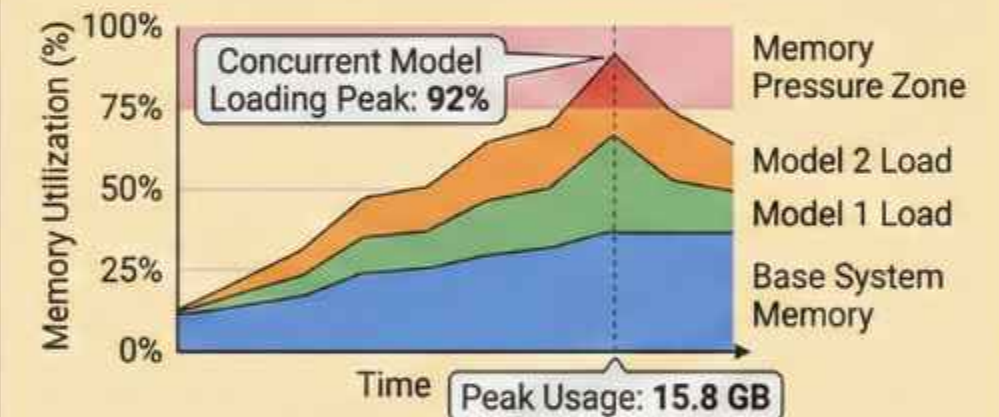
## Model Quantization



## Latency vs Quality



## Memory Stress Test



# INVESTIGATION & EXPLORATION



Memory Reduction:

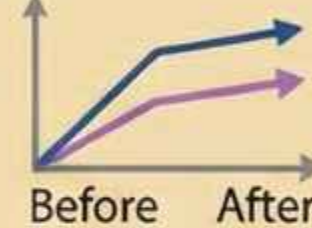
**40-50% ↓**

( $p < 0.01$ )



Latency Improvement:

**25-35% ↓**



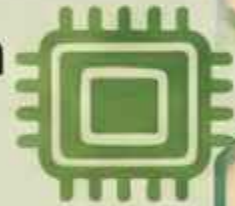
Efficiency Gains:

**2x / 15% ↓**



## Positive Performance Conclusions

- ✓ Significant memory footprint reduction
- ✓ Accuracy remains uncompromised
- ✓ Substantial efficiency improvements on all devices



## Nuanced Trade-offs & Observations

- ⚠ Latency improved, but extreme quantization
- ⚠ Minor accuracy drops in edge cases
- ⚠ Necessary trade-off for ultra-low power



## Efficiency Gains & Scalability

- ▲ Double the throughput on multi-core systems
- ✓ Reduced energy consumption



## Secondary Analytical Points

- 🔍 Memory bandwidth may be a bottleneck
- 🔍 Issue with larger batch sizes
- 🔍 Warrants further investigation



# TOOLS & TECHNOLOGIES

## Unified Technology Justification

### Frameworks



**PyTorch:** Flexible, dynamic deep learning research & development platform.



**TensorFlow:** Production-ready, scalable machine learning framework.



**FastAPI:** High-performance, modern, async web API framework.

### Models & Processing



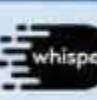
**TinyLlama:** Compact language model for efficient edge devices.



**ResNet FER:** Accurate facial expression recognition network.



**SER:** Speech Emotion Recognition for audio sentiment analysis.



**Faster-Whisper:** Fast, accurate speech-to-text transcription.

### Frontend & Storage



**HTML/CSS/JS:** Standard web technologies for responsive user interfaces.



**SQLite:** Lightweight, self-contained, serverless local storage.



**EdgeTTS:** Low-latency, high-quality text-to-speech synthesis.

### Reasoning

**Async Performance**  
Non-blocking operations,  
high throughput.



**Heterogeneous Model Support**  
Seamless integration of  
diverse AI models.

**Optimized Resource Utilization**  
Efficient CPU/GPU usage  
for edge constraints.



### Value



**Real-time Multimodal Interaction**

**Reduced Latency & Memory**

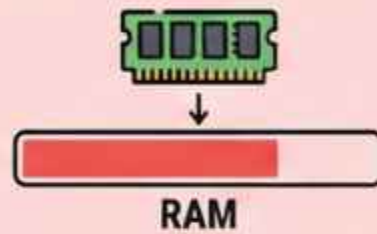


**Scalable Foundation**

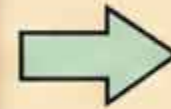
# CHALLENGES & SOLUTIONS



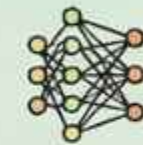
**Excessive RAM usage & silent crashes**



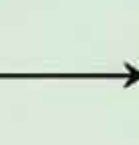
Leaks, Large Models, OOM Errors



**TFLite conversion, sequential loading, garbage collection**



Original Model



TFLite Conversion  
(Quantization, Optimization)



Sequential Loading  
(Chunking Data)

```
model = tf.lite.Interpreter(...);  
model.allocate_tensors();  
... process_chunk();  
gc.collect();
```

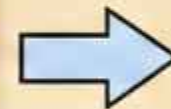
Garbage Collection  
(Explicit Memory Mgmt)



**Browser timeouts during LLM inference**



Long processing times, default limits



**Extended OS/network timeouts**



OS/Browser  
Settings

```
fetch(url, {  
  timeout: 60000  
});
```

Increase Timeout  
Limits  
(HTTP, Socket, Keep-Alive)

```
// Increase to 10 mins  
server.config.timeout = 600;
```

Backend/Server  
Config  
(Adjust Request Handling)

Asynchronous  
Processing  
(Background Tasks)



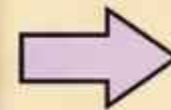
**TFLite tensor shape inconsistencies**



Expected Tensor  
(e.g., [1, 224, 224, 3])

Actual Tensor  
(e.g., [?, ?, 3])

Dynamic inputs, model conversion issues



**Custom TFLite tensor wrapper with dynamic allocation**



TFLite  
Interpreter  
Wrapper

```
wrapper = CustomInterpreter(model_path);  
wrapper.resize_tensor(input_shape);  
wrapper.invoke();  
output = wrapper.get_output();
```

Dynamic Shape  
Analysis  
(Inspect Input Data)

Runtime Resizing  
(ResizeInputTensor)

Validated Tensor  
Output  
(Consistent Shapes)

# GANTT CHART

